

Lab 05 – Networking Fundamentals with Cisco Packet Tracer

Device Configuration, Network Topology & Wireless Troubleshooting

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INTRODUCTION

This lab report documents a series of hands-on networking activities completed using Cisco Packet Tracer. The exercises covered foundational networking concepts including structured cabling, wireless device connectivity, command-line interface (CLI) configuration, packet analysis, network topology design, and centralized network management. Each activity was designed to build practical skills applicable to real-world network administration and troubleshooting.

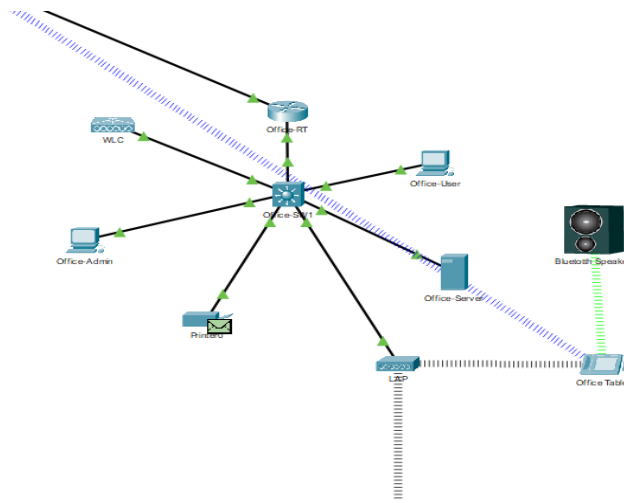
1.1.4 Packet Tracer – Create Realistic Structured Cabling in the Physical Workspace and Cabling Devices in a Rack



This activity focused on physical network infrastructure design within Cisco Packet Tracer's Physical Workspace. A patch panel was installed and wall-mount jacks were positioned throughout the simulated office environment. Network cables were then routed to interconnect all devices within the rack. Upon completion, each connection was tested and verified to be functioning correctly, reinforcing the importance of structured cabling standards in professional network deployments.

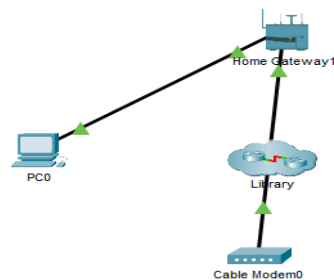
1.1.6 Packet Tracer – Connect Devices Using Wireless Technologies

2.0.3 Packet Tracer – Examine Packets in the Small Office



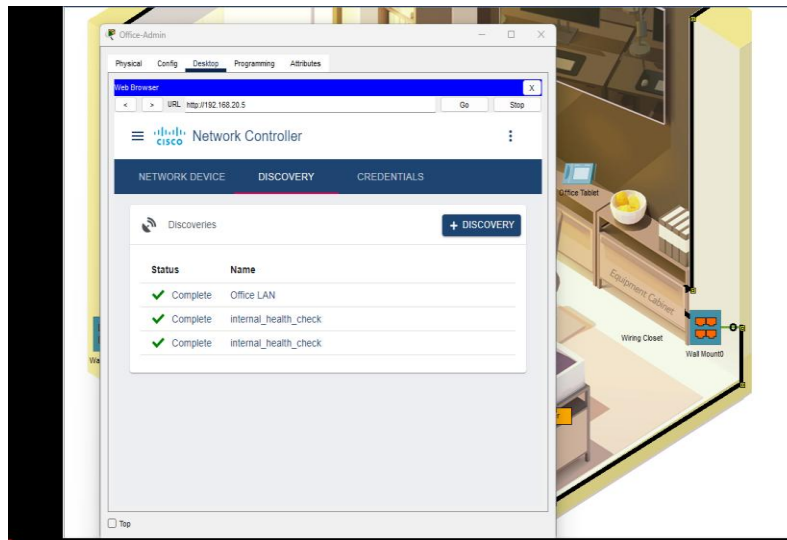
Using Packet Tracer's simulation mode, this activity examined the behavior of Protocol Data Units (PDUs) as they traverse a small office network. Both simple and complex PDUs were generated and tracked across network devices, allowing for detailed inspection of encapsulation, addressing, and forwarding decisions at each hop. The exercise reinforced an understanding of how data is packaged, routed, and delivered across layered network architectures.

2.0.7 Packet Tracer – Edit Topologies



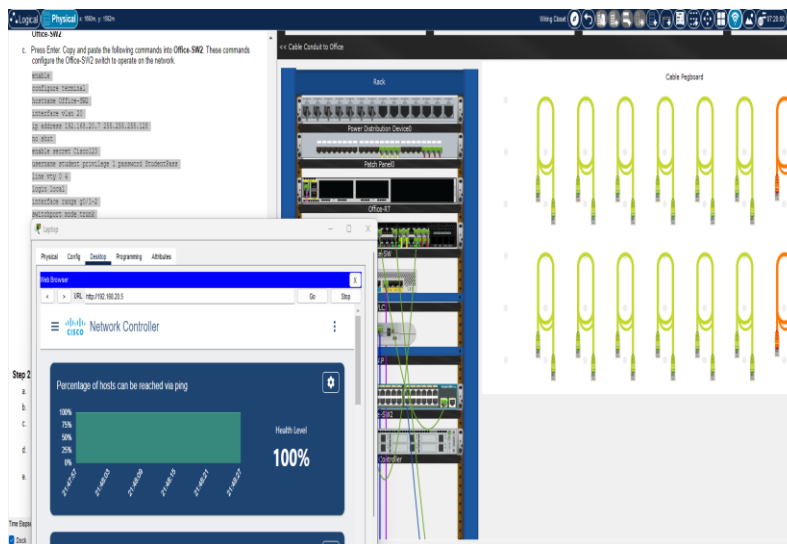
This activity focused on modifying an existing network topology within Cisco Packet Tracer. A second switch was added to the topology and a new PC was connected to it. Device clustering was explored as a method for organizing complex topologies, and a secondary home network cluster was configured. The activity developed practical skills in expanding and restructuring network diagrams to meet evolving infrastructure requirements.

2.1.3 Packet Tracer – Monitor Your Network Using a Network Controller



This activity introduced centralized network management through the Cisco Network Controller interface within Packet Tracer. The controller was configured and used to perform network discovery, identifying all connected devices across the topology. The exercise provided an introductory understanding of software-defined networking (SDN) principles and the operational advantages of managing distributed infrastructure through a unified management plane.

2.1.5 Packet Tracer – Manage and Configure Your Network Using a Network Controller



Building on the previous activity, this exercise extended the use of the Cisco Network Controller to actively manage and expand the simulated office network. Device discovery was performed to identify existing infrastructure, and a new network device was provisioned and integrated into the office topology through the controller interface. The activity demonstrated how centralized management tools streamline device onboarding and network expansion in enterprise environments.

2.2.1 Packet Tracer Tutored Activity – Troubleshoot a Wireless Connection

Congratulations!! You have successfully fixed the laptop wireless issues.

Summary

In this activity, you have accomplished the following learning objectives:

- ☒ How to identify a laptop wireless issue
- ☒ How to access a wireless router to see its configurations
- ☒ How to connect a laptop to wireless using WPA2-PSK

Reflection Question Answers

What methods did you use to determine which PC had connectivity issues?
Examples that you can use are ping, tracert, or you can navigate to a web page.
Possible answer

How did you discover the IP address of the wireless router?
From the PC command prompt, use the ipconfig command to determine the IP address of the default gateway.
Possible answer

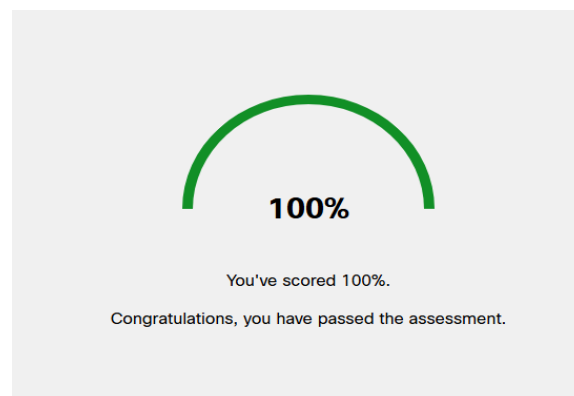
What is the difference between using the ping command and the tracert command? Which do you prefer?
The tracert command provides path information to the destination, but the ping command provide information about the destination.
Possible answer

Why is it important to change the default username and password for administrative access to a wireless router?
The default username and password is published by the manufacturer and readily available on the internet.
Possible answer

How do you feel about the process of troubleshooting a wireless network as a result of completing this activity?
Some people find the troubleshooting process to be complex and frustrating. Sometimes it seems it is hard to know where to get started. However, troubleshooting can be a very rewarding process because

This tutored activity presented a wireless connectivity fault scenario requiring systematic diagnosis and resolution. The affected laptop was identified as unable to connect to the wireless network. Using diagnostic tools including ping and tracert, the source of the issue was isolated to an incorrect wireless security configuration. Access to the wireless router's administrative interface was obtained via the ipconfig command to determine the default gateway IP address, and the connection was successfully restored by configuring the laptop to authenticate using WPA2-PSK. The activity highlighted the importance of methodical troubleshooting workflows and familiarity with wireless security protocols.

COURSE FINAL ASSESSMENT



The course final assessment evaluated cumulative knowledge across all module topics, including physical device setup, icon identification, wireless connectivity configuration, and network controller operation. The assessment was completed with a score of 100%, demonstrating proficiency across all examined competency areas and reinforcing confidence in navigating and operating within the Cisco Packet Tracer environment.

CONCLUSION

This lab provided a comprehensive introduction to core networking concepts through guided, hands-on simulation in Cisco Packet Tracer. Activities spanned physical cabling and rack configuration, wireless connectivity across multiple technologies, CLI-based device management, packet-level traffic analysis, topology design and modification, centralized network monitoring, and wireless fault diagnosis.

The primary challenge encountered during the lab was navigating device configuration through the CLI. Revisiting the provided tutorial materials was necessary to develop familiarity with the IOS interface and the location of relevant configuration options within the graphical environment. This challenge ultimately reinforced the value of reference documentation and structured learning resources in a technical context.

The skills developed through this lab have direct applicability to entry-level network administration and support roles. As organizations continue to rely on interconnected infrastructure — spanning wired, wireless, and software-defined networks — the ability to configure, monitor, and troubleshoot network devices represents a foundational and increasingly valuable technical competency.

REFERENCES

<https://www.netacad.com/launch?id=dc0847b7-e6fc-4597-bc31-38ddd6b07a2f&tab=curriculum&view=965aac98-b0c3-5c11-89ae-92f5e8b0a52f>
Cisco NetAcad – Packet Tracer course tutorials and activity modules.

Generative AI Searches:

N/A

Collaboration:

This lab was completed independently.